

NAGALAXMIK

Data Scientist

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SUMMARY

- Highly Motivated and accomplished MCA professional with over 10 years of experience, including 3 years specializing in Data Science and ML/AI, with expertise in Generative AI(Artificial Intelligence),EDA, Python, SQL, AWS, and Java
- Successfully deployed innovative healthcare chatbots, boosting user engagement by 25% using cutting-edge technologies like LangChain, Llama 3.1, FAISS, RAG, and Prompt Engineering.
- Proficient in AWS and MLOps, with proven skills in leveraging advanced analytics, predictive modeling, and data visualization to drive impactful business insights and operational efficiency.

PROFESSIONAL EXPERIENCE

2024

• Gen AI Prompt Engineer

BEPEC SOLUTIONS PVT. LTD.

Client and Project Name : Healthcare Chatbot using RAG, Lang chains, Faiss, and Llama3.1

Achievements:

1. Built a healthcare chatbot using Langchain to manage conversations and Faiss as a vector database to quickly retrieve medical information. The chatbot used Retrieval-Augmented Generation (RAG) to give more accurate and relevant answers to users' healthcare questions. I also created custom prompt templates to guide the chatbot's responses and used Amazon Bedrock to handle Llama3.1 large language model operations.
2. I deployed the chatbot using Streamlit, making it easy for healthcare professionals and patients to use.
3. This led to a 40% boost in response accuracy and user engagement increased by 25%, making the chatbot more reliable.

Tools\Technology:

- Python, Langchain, Llama 3.1, Amazon Bedrock, Faiss, Streamlit, Docker, AWS, RAG, Chatbots, Prompt Engineering, Healthcare AI, Model Deployment

2017 - 2018

• Data Scientist

Tata Consultancy Services

Client and Project Name : CBA - Digital Transformation

Project Summary :

The Commonwealth Bank(CBA) is one of Australia's leading financial institutions with businesses in New Zealand, Asia and the United Kingdom.This project was to perform focused digital marketing by predicting the potential customers who will convert from liability customers to assets customers.

Achievements:

1. Conducted data preprocessing and exploratory data analysis (EDA) using the python libraries: Matplotlib,Seaborn.
2. Developed predictive models, including Logistic Regression, Support Vector Machine, and Random Forest Classifier.
3. Evaluated the models with precision, recall, and AUC metrics, we improved efficiency in identifying, Potential customers, achieving an impressive AUC score of 0.97 through iterative refinement.
4. Conducted a proof of concept to devise and execute automated pipelines for deploying ML models, across various environments, utilizing containerization technologies (such as Docker, Kubernetes) and Orchestration tools (like Kubeflow) into AWS.

Tools\Technology:

- Python,Numpy,Pandas,Matplotlib, Seaborn,Sklearn,pickle,CI/CD,Kubernetes,AWS

PROFESSIONAL EXPERIENCE

2016 - 2017

- Data Scientist

Tata Consultancy Services

Client and Project Name: CBA - Loan Eligibility Prediction

Project Summary:

The Commonwealth Bank(CBA) is one of Australia's leading financial institutions with businesses in New Zealand, Asia and the United Kingdom. This project to predict if an applicant should get a loan or not by building classification model.

Achievements :

1. Data quality was ensured by rectifying missing values, addressing data skewness, and mitigating Multicollinearity using Python libraries such as Pandas, Seaborn, and Matplotlib.
2. Developed and trained predictive models such as XGBoost and Random Forest models to evaluate, the risk associated with each loan applicant based on their financial history, credit score, and other relevant features.
3. Established Continuous Integration and Delivery (CI/CD): Developed CI/CD pipelines to automate the testing, validation, and deployment of ML models

Tools\Technology:

- Python,Numpy,Pandas ,matplotlib,seaborn Sckit_learn,xgboost.

2015 - 2016

- Junior Data Analyst

Tata Consultancy Services

Client and Project Name : Tata Communications Limited

Project Summary :

Tata Communications Limited, a worldwide leader in telecommunications and digital infrastructure within the Tata Group, focuses on predicting Customer Lifetime Value (CLTV). The CLTV metric gauges the total anticipated revenue a customer is likely to generate throughout their association with the company. This initiative aims to optimize sales and pricing strategies, enhance customer satisfaction, and mitigate churn rates.

Achievements:

- 1.Performed exploratory data analysis (EDA) to understand the data and identify features that contribute to CLTV based on the aforementioned factors.
- 2.Applied analytics techniques such as outlier detection, quantiles hypothesis testing(One-Sample t-test, Paired Samples t-test)

Tools\Technology:

- Python,Numpy,Pandas,Matplotlib,Seaborn,Scikit-learn,MYSQL.

2008 - 2018

- Assistant Consultant

Tata Consultancy Services

- Worked at TCS, Bangalore, from June 2008 to October 2018, with an added focus on data science roles from 2015 to 2018.

TECH STACK

Gen AI	RAG	Lang chains(LLM)	Llama3.1	NLP	Python	PyTorch	Tensorflow	Numpy
Pandas	Matplotlib	Scikit-learn	Seaborn	Machine Learning				
Supervised Learning algorithms(ML algorithms)				Unsupervised Learning algorithms			Statistics	
Statistical Analysis		MySQL	Tableau	MS-Excel	Java	Docker	Kubernetes	MLOPS
MLflow	KubeFlow	AWS Cloud	AWS Databricks	AWS SageMaker		Azure	GCP	Git&GitHub
Agile								

TRAINING AND PROHIBITION PERIOD

- 2023 - Present** • **Data Scientist - DSML AUG 2023**
Scaler Academy
Gaining expertise in AI/ML development (Data Science and Machine Learning) through training at Scaler Academy
- 2022 - 2023** • **Data Science Intern**
ProjectPro
To further my engagement in data science, I've successfully completed a program with ProjectPro during a career break
- 2018 - 2022** • **Career break for childcare**
Took a career break from 2018 to 2022 to prioritize caring for my child, resulting in a software gap in my professional journey.

EDUCATION

- 2001 - 2004** • **MCA (Master of Computer Applications)**
Andhra University
- 1998 - 2000** • **BSC**
MPR Degree College